

PROSPER PLATFORM

Status Report & Roadmap to First Real Client

Fintech tokenized-yield platform built on Stellar — Internal Backoffice, Client / Partner Portal and public Developer Portal. This document summarises what is built, what is demo vs real, and what is required to onboard a production client.

1. What the platform is

Regulated tokenized-yield infrastructure with three surfaces:

- **Backoffice** — for internal Prosper staff (ops, compliance, finance, super-admin).
- **Client / Partner Portal** — for partner organisations that invest and manage their own end-customers.
- **Developer Portal** — public docs + OpenAPI for technical integrators.

Stack: FastAPI (async) · MongoDB · React + Shadcn UI · Phosphor Icons · Recharts. Brand match with `prosper.foundation` (light + dark, IBM Plex / Chivo, Swiss-style "Control Room" aesthetic).

2. What is built

Backend (~50 endpoints)

- Emergent Google OAuth + 7-day sessions + seeded demo tokens for automation.
- Multi-tenant scoping by `org_id`; `@prosper.foundation` users auto-flagged as super-admin.
- Operations Queue (4-eyes / two-signer): mint and critical actions create an `apv_*` that a different user must approve.
- MFA TOTP (setup / verify / disable) with challenge at approval time.
- HMAC-SHA256 webhook signatures (`X-Prosper-Signature: t=..., v1=...`), delivery log and auto-retries.
- Idempotency-Key middleware on every write mutation.
- APScheduler background workers (webhook retries, SLA checks, EOD NAV snapshots).
- Prosper Stellar upstream proxy (`prosper_client.py`) — code-ready, mocked today because the egress is blocked.
- Upstream diagnostic GET `/api/admin/prosper-upstream` (enabled / reachable / authenticated / error).
- KYC / KYB Object Storage with async upload (`httpx.AsyncClient`), magic-byte sniffing, soft-delete.
- Global **■**K search across orgs, transactions, positions, API keys, onboarding.

Frontend (33 screens)

- **19 Backoffice screens:** Dashboard, Clients, ClientDetail (with docs), Onboarding, Compliance, Funds, Products, Positions, Treasury, Transactions, Reconciliation, Integrations (with upstream card), ApiKeys, Webhooks, Alerts, Reports, UsersAdmin, AuditLog, Approvals.
- **13 Client Portal screens:** Overview, Organization (with docs), Users, Balances, Transactions, Yield, EndCustomers, Integrations, ApiKeys, Webhooks, Reports, Compliance, Settings.
- **Public:** Login + Developer Portal /developers (hero, quickstart, endpoint reference with sidebar, curl/js/python examples, /openapi.json link).
- **■K command palette** · Alerts bell · CSV exports · date-range pickers · pagination.
- Landing quick-demo links → Backoffice / Portal / Dev Portal (no OAuth, seeded tokens).

Seed data (auto on first boot)

5 organisations · 2 funds (PROS prod + sandbox) · 5 products · 35 positions · 120 transactions · 120 reconciliation records · 6 API apps · 3 API keys · 3 webhooks with 18 deliveries · 4 alerts · 12 reports · 40 audit events · 24 end customers · 2 demo users with fixed tokens (super_admin + client_admin).

Tests

49/49 pytest green covering auth, dashboard, organisations, onboarding, compliance, funds, products, positions, treasury, transactions, reconciliation, integrations, alerts, reports, audit, users, documents (upload/list/download/delete/magic-bytes/cross-org 403) and the upstream diagnostic. Four iterations of the testing agent confirmed end-to-end flows.

3. What is DEMO vs REAL today

■ 100% DEMO (tagged `is_demo: true`)

All seed data is auto-generated on the first backend boot and tagged so it can be wiped surgically via POST `/api/admin/wipe-demo`.

Collection	Reality
Organisations	5 (Alemany, Beachside, Quirón, Meridian, Andes) — invented
Funds	2 (PROS prod + PROS sandbox) — NAV / supply / treasury_address simulated
Products	5 (term staking 30/90/180/365d + liquid) — hardcoded APRs
Positions	35 — principal and accrued interest random
Transactions	120 — tx_hash = SHA256(prosper_tx_id), never hit Stellar
Reconciliation	120 — on/off-chain match computed, not verified on chain
API apps / keys / webhooks	6 / 3 / 3 — no real traffic
Alerts / Reports / Audit	4 / 12 / 40 — timestamps relative to boot
End customers	24 — fake names and emails
Demo users + sessions	2 (super_admin + client_admin) — fixed tokens for automation

■ REAL today (platform layer)

- **Backend / frontend code** — production-grade.
- **Google OAuth (Emergent)** — a real production user gets a real session, separate from demo tokens.
- **Object Storage (Emergent)** — uploaded KYC files persist for real, not in memory.
- **MongoDB** — real collections (no custom index tuning yet).
- **APScheduler workers** — scheduler is actually running every minute.
- **HMAC webhooks** — signatures are computed and retried; destination URLs are demo (httpbin etc.).
- **Approvals two-signer + MFA TOTP + idempotency + audit log** — all logic is real; only the data flowing through is demo.

■ MOCKED (code ready)

PROSPER STELLAR UPSTREAM

Calls to the real Prosper Stellar rail (`prosper_client.py`). If `PROSPER_API_ENABLED=false` (current state) → simulated response, tx_hash = SHA256. Once `PROSPER_API_ENABLED=true` and the egress is open, `/v1/users/deposit`, `/withdraw` and `/Auth/Login` are already wired. Affected endpoints: POST `/api/positions/subscribe`, `/positions/{id}/redeem`, `/transactions/mint`.

4. What is missing to onboard a real client

Split into what **still needs work** vs what **already exists and only needs switching on**.

4.1 Identity & access

■ Done

- Roles client_admin / client_user / developer / viewer.
- Scoping by org_id, MFA TOTP, session expiry, audit log.

■ Missing

- Email allowlist / corporate-domain auth — today any Gmail can authenticate.
- Automatic email → org_id mapping on first login.
- Real invite flow: POST /api/users/invite exists, but does NOT send email. Connect SendGrid or Resend.
- Password-less / enterprise SSO (SAML / Okta) — P2.

4.2 Onboarding (KYB / KYC) end-to-end

■ Done

- Cases POST /api/onboarding with status, SLA 48h, progress %, risk_score.
- Review queue in Backoffice → Compliance.
- PDF / image upload with magic-byte validation, scoped by org.
- Full audit log of every decision.
- Optionally pluggable into the two-signer Approvals queue.

■ Missing

- Public-facing onboarding screen for clients to apply *before* having an account.
- Real Sanctions / PEP / Travel-rule provider (ComplyAdvantage, Sumsub, Chainalysis KYT) — fields exist but values are hardcoded pending.
- Tax forms (W-8 / W-9 / CRS) — schema does not exist yet.
- Document checklist per applicant type (individual / corporate / trust).
- E-sign of contracts (DocuSign / SignWell).

4.3 Treasury & client funds

■ Done

- Subscribe / Redeem with prosperTxId as idempotency anchor.
- On-chain ↔ off-chain reconciliation with manual resolution.
- Treasury console (issuer, treasury, reward_pool, fee accounts).

■ Missing

- Wire the real Stellar rail (depends only on infra egress → 1 line of .env).
- Client wallet flow: today stellar_address comes from the proxy mock; in production the client either provides one or Prosper issues a custodial wallet.
- USDC fiat on/off ramp — Circle, dLocal, Lemon, Bitso. Model does not exist yet.

- Per-org limits (daily / monthly subscribe / redeem cap) — schema field missing.

4.4 Product / fund operations

■ Done

- Funds CRUD, products CRUD, NAV snapshots history, position tracking.

■ Missing

- Automated NAV publishing — the job runs but values are mocked.
- Real yield accrual cron (compute accrued_interest per position from the product APR).
- Payout / claim UI — endpoint exists in transactions, no UI yet.

4.5 Daily operations (Backoffice)

■ Done

- Dashboard, alerts, audit log, reconciliation queue, approvals queue, ■K search, CSV exports.

■ Missing

- Assignable tasks (assignee, deadline) — schema prosper_tasks exists, no UI.
- Notifications (email / Slack on pending mint, KYC, critical alerts).
- Internal status page (webhook p95, recon lag, scheduler health).

4.6 Regulatory compliance

■ Missing

- Regulatory reports (CNV Argentina, UIF, FATCA) — generic Reports model exists, specific formats to be coded.
- Retention policies (purge audit logs / sessions / idempotency by window).
- Per-client data export (right to portability).

4.7 Production / infra

■ Missing

- Network egress to the Prosper upstream (single infra action today).
- Transactional email (SendGrid / Resend) for invites + notifications.
- MongoDB backups + replica set.
- MongoDB indexes (org_id, prosper_tx_id, created_at, status) for scale.
- Rate limiting per API key.
- Observability (Sentry / Datadog).

5. Shortest path to the first real client

If the goal is to onboard, say, Alemany Capital for real, the priorities are:

#	Action	Effort	Outcome
1	Open egress + PROSPER_API_ENABLED=true	Infra — 0 days of code	Real subscribe / redeem / mint.
2	Transactional email (Resend) + functional invite flow	1 day	Real users joining their org via signed link.
3	Real KYC / KYB provider (Sumsub or ComplyAdvantage)	2–3 days	Replace pending mocks with sanctions / PEP / KYC results.
4	Public onboarding screen	1–2 days	Clients can apply before having an account.
5	Domain allowlist in auth callback	Half a day	Only approved corporate domains can authenticate.
6	Wipe demo + create real Alemany org	Minutes	Existing endpoint.

BOTTOM LINE

The platform is demo-ready and end-to-end tested. With those 4–5 blocks shipped, the first real partner can apply, be reviewed, be approved, perform their first real subscribe on Stellar, see their position in the portal and receive signed webhooks — without further platform changes.